UBER Data Analyst Project Repor

1. Project Overview

This project focuses on analyzing ride booking data for July 2025 to derive actionable insights related to operations, customer behavior, and revenue performance. SQL and Power BI were used to process, analyze, and visualize the data.
The primary objective of this analysis is to support data-driven decision-making for a ride-hailing platform by identifying trends in booking success, cancellations, vehicle usage, payment methods, and customer satisfaction.

2.Business Problem

Ride-hailing platforms like Uber operate in a highly competitive and dynamic environment where customer satisfaction, operational efficiency, and revenue optimization are critical. Despite having large volumes of ride booking data, the business faces challenges in understanding key performance drivers such as ride success rates, cancellation patterns, customer and driver behavior, vehicle category performance, and preferred payment methods.

Without structured analysis, it is difficult for stakeholders to identify peak demand periods, reduce cancellations, improve driver allocation, and enhance customer experience. Therefore, the business requires a data-driven approach to transform raw booking data into meaningful insights that can support strategic and operational decision-making.

3. STAR Method Explanation

Situation

The ride-hailing business generates a large volume of booking data on a daily basis. However, raw data alone does not provide meaningful insights into business performance, customer behavior, cancellation patterns, or revenue trends.

Task

The task was to analyze booking data using SQL and visualization tools to answer key business questions related to:

- **Ride success and failure rates**
- **Customer and driver ratings**
- **Vehicle category performance**
- **Cancellation reasons**
- **Revenue and payment method trends**

Action

- **Imported and explored the dataset to understand structure and data quality**
- **Cleaned and validated the data for accurate analysis**

- **Wrote SQL queries to extract and aggregate insights**
- **Analyzed booking status, ride distance, ratings, cancellations, and payment methods**
- **Designed interactive Power BI dashboards to visualize trends and patterns**
- **Interpreted insights from both SQL outputs and dashboard visuals**

Result

The analysis identified peak ride demand periods, high-performing vehicle types, dominant digital payment methods, and key cancellation reasons. These insights can help improve operational efficiency, optimize driver allocation, reduce cancellations, and enhance overall customer satisfaction.

4. Dataset Description

- **Time Period: July 2025**
- **Records: Ride booking–level transactional data**
- **Key Columns:**
  - **Booking_ID,Booking_Status**
  - **Booking_Value**
  - **Customer_ID**
  - **Driver_ID**
  - **Pickup_Location**
  - **Drop_Location**
  - **Ride_Distance**
  - **Ride_Time**
  - **Vehicle_Type**
  - **Payment_Method**
  - **Customer_Rating**
  - **Driver_Rating**

5. Tools & Technologies

- **SQL (POSTGRSQL): Data extraction, filtering, aggregation, and analysis**
- **Power BI: Interactive dashboards and visual insights**

- **Microsoft Excel: Initial data review and validation**

6. Data Analysis using SQL

We perfomed structred analysis in PostreSQL to answer key business question:

1. Retrieve all successful bookings:

SELECT * FROM ridess_data

WHERE Booking_Status = 'Success'

	date date	time time without time zone	booking_id character varying (30)	booking_status character varying (30)	customer_id character varying (30)	vehicle_type character varying (30)	pickup_location character varying (255)	drop_location character varying (255)
1	2024-07-...	22:20:00	CNR2940424040	Success	CID225428	Bike	Magadi Road	Varthur
2	2024-07-...	19:59:00	CNR2982357879	Success	CID270156	Prime SUV	Sahakar Nagar	Varthur
3	2024-07-...	09:02:00	CNR1797421769	Success	CID939555	Mini	Rajajinagar	Chamarajpet
4	2024-07-...	04:42:00	CNR8787177882	Success	CID802429	Mini	Kadugodi	Vijayanagar
5	2024-07-...	09:51:00	CNR3612067560	Success	CID476071	Bike	Tumkur Road	Whitefield
6	2024-07-...	23:33:00	CNR4787583516	Success	CID923404	Prime Plus	Hosur Road	Jayanagar
7	2024-07-...	04:03:00	CNR7943634301	Success	CID647026	Prime Plus	Kammanahalli	Rajajinagar
8	2024-07-...	13:18:00	CNR4524472111	Success	CID540929	Auto	Cox Town	Yelahanka

Total rows: 44271 Query complete 00:00:00.454 CRLF Ln 5, Col 1

2. Average ride distance by vehicle type

SELECT Vehicle_Type, AVG(Ride_Distance)

as avg_distance FROM ridess_data

GROUP BY Vehicle_Type

	vehicle_type character varying (30)	avg_distance numeric
1	eBike	15.6303183721156655
2	Auto	6.2099626645706426
3	Bike	15.7466889041499068
4	Prime Sedan	15.7126705653021442
5	Prime Plus	15.3720267348142324
6	Mini	15.5738244983528002
7	Prime SUV	15.2022160664819945

3.Cancellation analysis by customers .

SELECT COUNT(*) FROM ridess_data

WHERE Booking_Status = 'cancelled by Customer'

	count bigint
1	0

4. List the top 5 customers who booked the highest number of rides:

SELECT Customer_ID, COUNT(Booking_ID) as total_rides

FROM ridess_data

GROUP BY Customer_ID

ORDER BY total_rides DESC LIMIT 5

	customer_id character varying (30)	total_rides bigint
1	CID340854	4
2	CID463543	3
3	CID657000	3
4	CID356460	3
5	CID896927	3

6. Find the maximum and minimum driver ratings for Prime Sedan bookings:

SELECT MAX(Driver_Ratings) as max_rating, MIN(Driver_Ratings)

as min_rating FROM

ridess_data

WHERE Vehicle_Type = 'Prime Sedan'

	max_rating numeric	min_rating numeric
1	5.00	3.00

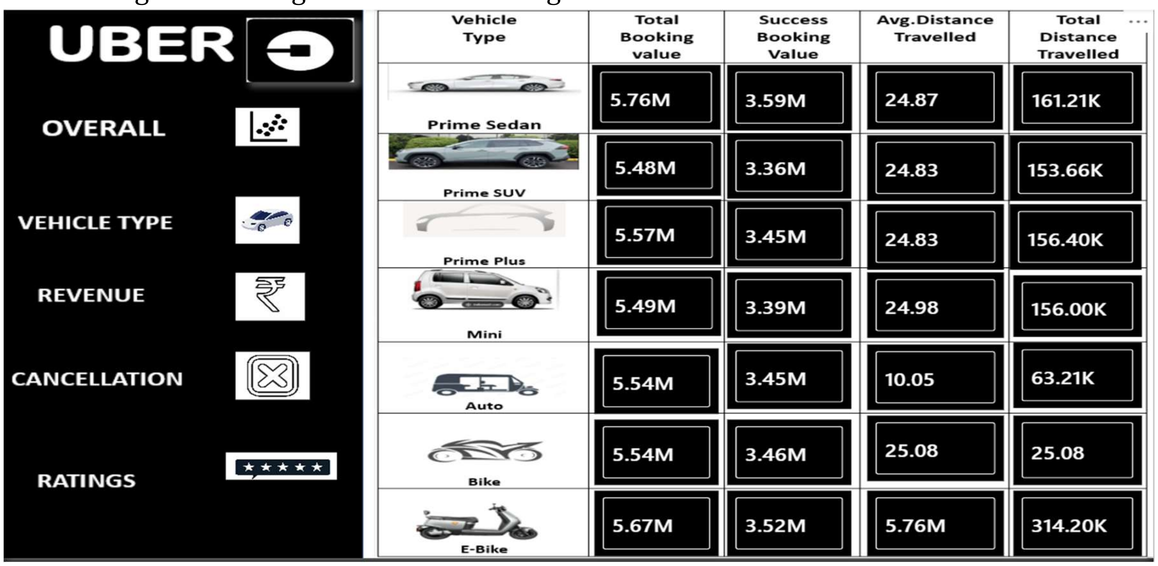
7. Power BI Dashboard Analysis

Power BI dashboards were created to visualize trends and patterns effectively. Interactive visuals helped identify daily ride trends, revenue distribution, and performance metrics.

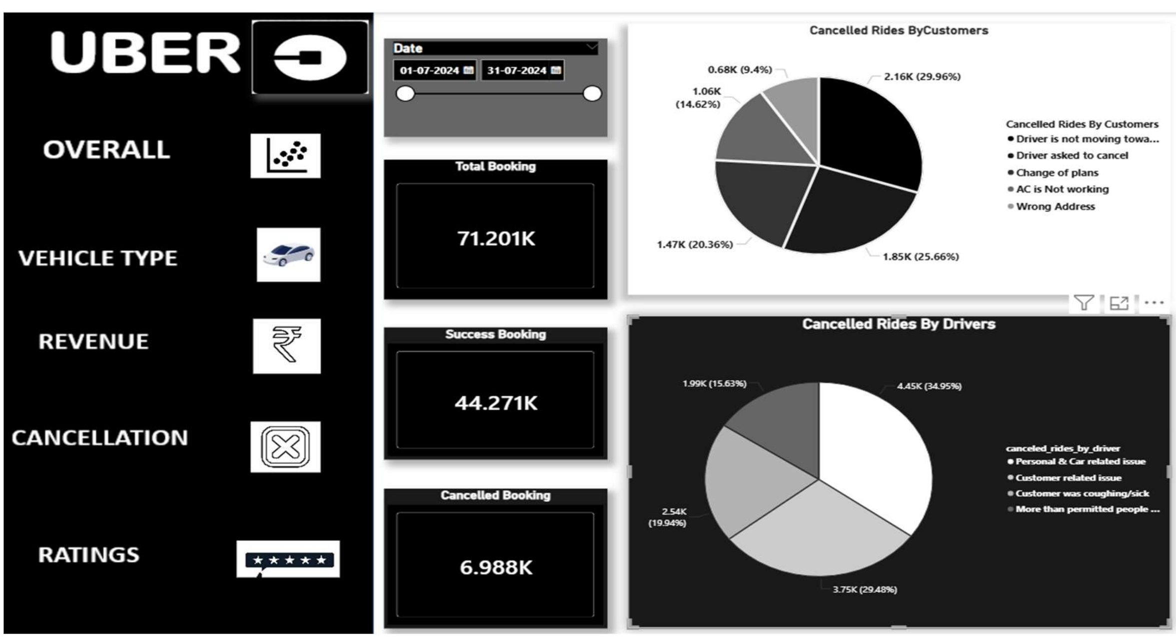
* Power BI Insights

- Ride volume peaked during weekends.

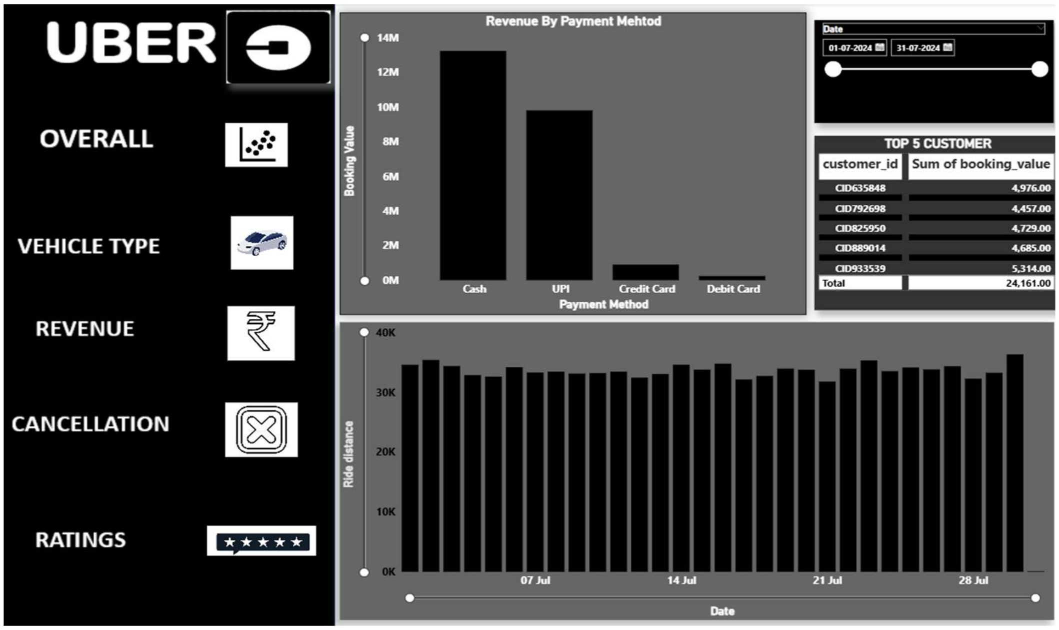
- vehicles generated higher Success Booking



- Lower ratings were often associated with cancelled rides by customers



- Digital payment methods dominated overall transactions.



8. Business Recommendations

- Improve driver availability during peak hours.
- Reduce cancellations through better driver-customer matching.
- Promote high-performing vehicle categories.
- Incentivize drivers to maintain high ratings.

9. Conclusion

This project analyzed July 2025 ride booking data using SQL and Power BI to derive actionable business insights. Key findings included peak demand trends, high-performing vehicle types, preferred payment methods, and major cancellation reasons.
The analysis highlighted the relationship between ride completion, ratings, and revenue performance.
Overall, the project demonstrates strong data analysis and visualization skills to support data-driven decision-making.